ADVENTURES IN BIG MATH · 3RD GRADE MATH

MISSION 03 OF 08 · WORKSHEET PACK

Number Power

Every graded set, every week — 4 weeks, 20 two-page sheets, with an answer key after each week.

WEEK 1	WEEK 2	WEEK 3	WEEK 4
5 sets	5 sets	5 sets	5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

How Big Is a Million

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Read the place, then say what the digit is worth. Type digits only — no commas needed.

Warm-Up

11 ITEMS · RECALL

1 How many tens in 100?

2 How many hundreds in 1,000?

3 10 × 100

4 100 × 100

5 1,000 × 10

6 How many tens in 50

7 How many hundreds in 700

8 10 × 10

9 100 × 10

10 Zeros in ten thousand

11 How many tens in 1,000

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. Write six million forty thousand five (6,040,005 — hold the empty places with zeros) ....

13. The value of the 5 in 5,400 ....

14. The value of the 8 in 380 ....

★ Challenge

NOT GRADED · REASON IT OUT

15. How many thousands in 10,000?

16. In 3,472,861 — what is the 4 worth? (It sits in the hundred-thousands place)

17. In 3,472,861 — what is the 7 worth? (Ten-thousands place)

18. In 5,208,043 — what is the 2 worth? (Count the places from the right)

19. How many thousands make 4,000,000? (4,000 × 1,000 = 4,000,000)

20. How many zeros in one million? (1,000,000)

21. 1,000 × 1,000 (A thousand thousands)

22. The value of the 2 in 24,000

23. Thousands in 250,000

24. Hundreds in 45,000

25. Millions in one billion

26. Zeros in a hundred million

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Ten Times Bigger

Multiplying by ten shifts every digit one place left. Watch the zeros.

Warm-Up

9 ITEMS · RECALL

1 40 × 10

2 700 × 10

3 60 × 100

4 8 × 1,000

5 3,200 ÷ 10

6 20 × 10

7 500 × 10

8 90 × 100

9 4,000 ÷ 10

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

10. 230 × 100 (Two more zeros)

11. 9,000 ÷ 100 (Take two zeros away)

12. 350 × 100

13. 1,200 × 10

14. 5,600 ÷ 10

Challenge

NOT GRADED · REASON IT OUT

15. 45,000 ÷ 1,000
16. 56 × 1,000 (Three more zeros)
17. 740,000 ÷ 10,000 (Four zeros off)
18. How many times bigger is 800,000 than 800? (Three places left)
19. The 6 in 6,000 is worth how many times the 6 in 60? (Two places apart)
20. How many 10,000s make one million? (1,000,000 ÷ 10,000)
21. 7 × 100
22. 60,000 ÷ 1,000
23. 84,000 ÷ 100
24. 25 × 1,000
25. The 3 in 30,000 is worth how many times the 3 in 300
26. How many 1,000s make ten million

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

NAME __________

DATE __________

RIGHT ____ OF ____ TRIED

Compare, Order, Round

Rounding depends on which place you were asked for. Read the question twice.

Warm-Up

12 ITEMS · RECALL

1 Round 47 to the nearest ten	2 Round 83 to the nearest ten	3 Round 65 to the nearest ten	4 Round 128 to the nearest ten
5 Round 254 to the nearest ten	6 Round 999 to the nearest ten	7 Round 34 to the nearest ten	8 Round 76 to the nearest ten
9 Round 55 to the nearest ten	10 Round 245 to the nearest hundred	11 Round 380 to the nearest hundred	12 Round 909 to the nearest ten

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Round 2,499 to the nearest hundred (Look at the tens digit only) ....

14. Round 3,472 to the nearest hundred ....

15. Round 6,509 to the nearest thousand ....

★ Challenge

NOT GRADED · REASON IT OUT

16. Round 47,382 to the nearest thousand (382 down beats 618 up)

17. Round 47,382 to the nearest ten thousand (Now the ends are 40,000 and 50,000)

18. Round 863,209 to the nearest hundred thousand (63,209 up to 900,000 is the shorter trip)

19. Larger number: 408,916 or 480,169? Type it. (Compare the ten-thousands place)

20. Largest number that rounds to 5,000 at the nearest thousand (One less than the halfway point up)

21. Smallest number that rounds to 5,000 at the nearest thousand (Exactly halfway rounds up)

22. Round 24,650 to the nearest thousand

23. Round 89,999 to the nearest ten thousand

24. Round 12,345 to the nearest hundred

25. Largest whole number that rounds to 700 at the nearest hundred

26. Smallest whole number that rounds to 700 at the nearest hundred

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

Order of Operations

Build the groups first, then add and subtract what's left loose.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

12 ITEMS · RECALL

1 $3 + 4 \times 2$	2 $10 - 2 \times 3$	3 $12 \div 4 + 5$	4 $6 + 6 \div 2$
5 $2 \times 3 + 4 \times 5$	6 $20 - (4 + 6)$	7 $5 + 2 \times 3$	8 $8 - 3 \times 2$
9 $16 \div 4 + 1$	10 $9 + 9 \div 3$	11 $(5 + 5) \times 2$	12 $30 - (5 + 5)$

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $5 + 3 \times 4$ (Multiply first: $3 \times 4 = 12$) ....

14. $(5 + 3) \times 4$ (Brackets overrule everything) ....

15. $18 \div (3 + 6)$ ($3 + 6 = 9$ first) ....

16. $40 - 6 \times 5 + 2$ ($40 - 30 + 2$) ....

17. $3 \times (12 - 4) \div 6$ ($3 \times 8 = 24$, then $\div 6$) ....

18. $4 \times 5 - 6 \div 2$ ....

19. $(12 + 8) \div 4$ ....

20. $7 \times (3 + 2)$ ....

21. $48 \div (2 \times 3)$ ....

★ Challenge

NOT GRADED · REASON IT OUT

22. $2 + 3 \times 4 - 6 \div 2$ (2 + 12 - 3)

23. $(8 + 4) \times (9 - 6)$ (12 × 3)

24. $100 - 5 \times 12$

25. $(6 + 4) \times (8 - 3)$

26. $5 + 4 \times 3 - 12 \div 4$

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

Powers & Expression Duel

Enrichment day. The little raised number counts copies, not multiples.

NAME __________

DATE __________

RIGHT ____ OF ____ TRIED

CHALLENGE AND THE ERROR JOURNAL ARE ON THE BACK

Challenge

NOT GRADED · REASON IT OUT

1. 2^3 (2 × 2 × 2)
2. 5^2 (5 × 5)
3. 10^4 (A 1 with four zeros)
4. $3^2 + 4^2$ (9 + 16)
5. $2^5 - 2^3$ (32 - 8)
6. Bigger value: 4^3 or 3^4? Type it. (64 against 81)
7. $10^6 \div 10^3$ (Six zeros take away three)
8. How many zeros in 10^8? (The exponent counts them)
9. 3^3
10. 4^2
11. 5^3
12. 2^5

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

TEACHER PAGE · ANSWER KEY

Week 1 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

1.1 · How Big Is a Million

OUT OF 14

Warm-Up · 1–11 · 10 | 10 | 1000 | 10000 | 10000 | 5 | 7 | 100 | 1000 | 4 | 100

Core · 12–14 · 6040005 | 5000 | 80

Challenge · 15–26 · 10 | 400000 | 70000 | 200000 | 4000 | 6 | 1000000 | 20000 | 250 | 450 | 1000 | 8

1.2 · Ten Times Bigger

OUT OF 14

Warm-Up · 1–9 · 400 | 7000 | 6000 | 8000 | 320 | 200 | 5000 | 9000 | 400

Core · 10–14 · 23000 | 90 | 35000 | 12000 | 560

Challenge · 15–26 · 45 | 56000 | 74 | 1000 | 100 | 100 | 700 | 60 | 840 | 25000 | 100 | 10000

1.3 · Compare, Order, Round

OUT OF 15

Warm-Up · 1–12 · 50 | 80 | 70 | 130 | 250 | 1000 | 30 | 80 | 60 | 200 | 400 | 910

Core · 13–15 · 2500 | 3500 | 7000

Challenge · 16–26 · 47000 | 50000 | 900000 | 480169 | 5499 | 4500 | 25000 | 90000 | 12300 | 749 | 650

1.4 · Order of Operations

OUT OF 21

Warm-Up · 1–12 · 11 | 4 | 8 | 9 | 26 | 10 | 11 | 2 | 5 | 12 | 20 | 20

Core · 13–21 · 17 | 32 | 2 | 12 | 4 | 17 | 5 | 35 | 8

Challenge · 22–26 · 11 | 36 | 40 | 50 | 14

Fri · Powers & Expression Duel

ENRICHMENT

Challenge · 1–14 · 8 | 25 | 10000 | 25 | 24 | 81 | 1000 | 8 | 27 | 16 | 125 | 32 | 36 | 64

Rounding to Any Place

Find the place, look one to its right, decide.

NAME __________

DATE __________

RIGHT ____ OF ____ TRIED

Warm-Up

12 ITEMS · RECALL

1 Round 48 to the nearest ten	2 Round 43 to the nearest ten	3 Round 250 to the nearest hundred	4 Round 149 to the nearest hundred
5 Round 75 to the nearest ten	6 Round 612 to the nearest ten	7 Round 62 to the nearest ten	8 Round 67 to the nearest ten
9 Round 340 to the nearest hundred	10 Round 360 to the nearest hundred	11 Round 95 to the nearest ten	12 Round 450 to the nearest hundred

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Round 4,829 to the nearest hundred ....

14. Round 4,829 to the nearest thousand ....

15. Round 8,950 to the nearest hundred ....

16. Round 5,432 to the nearest thousand ....

17. Round 5,632 to the nearest thousand ....

18. Round 7,777 to the nearest hundred ....

Challenge

NOT GRADED · REASON IT OUT

19. Round 27,500 to the nearest thousand
20. Round 96,412 to the nearest ten thousand
21. A number rounds to 300 to the nearest hundred. Its smallest possible value
22. And its largest possible whole value
23. Round 18,750 to the nearest thousand
24. Round 46,499 to the nearest thousand
25. A number rounds to 8,000 to the nearest thousand — its smallest whole value
26. ... and its largest whole value

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

Estimate Before You Compute

Round first. Then the real answer has something to be checked against.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up 11 ITEMS · RECALL

1 Estimate 48 + 31 by rounding both to tens

2 Estimate 62 – 19

3 Estimate 19 × 21

4 Estimate 81 ÷ 9

5 Estimate 297 + 104

6 Estimate 52 + 48

7 Estimate 39 + 42 by rounding to tens

8 Estimate 71 – 28

9 Estimate 29 × 11

10 Estimate 63 ÷ 8

11 Estimate 198 + 205

Core GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. Estimate 4,812 + 3,190 to the nearest thousand

13. Estimate 612 – 388 to the nearest hundred

14. Estimate 3,964 ÷ 4

15. Estimate 486 + 317 to the nearest hundred

16. Estimate 4,120 – 1,890 to the nearest hundred

17. Estimate 396 × 21

18. Estimate 5,940 ÷ 6

19. Estimate 78 × 41

★ Challenge

NOT GRADED · REASON IT OUT

20. Estimate 187 × 4

21. True value of 187 × 4

22. Estimate the cost of 19 items at $4.95 each

23. Someone estimates 512 × 6 as 300. Type the real answer

24. Estimate 52 × 9

25. Estimate 12 items at $19.95

26. Someone estimates 407 × 8 as 320. Type the real answer

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

How Wrong Is the Estimate

An estimate you never compare to anything teaches you nothing.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

10 ITEMS · RECALL

1. Estimate of 48 + 31
2. True value of 48 + 31
3. The difference
4. True value of 19 × 21
5. Estimate of 19 × 21
6. Estimate of 39 + 42
7. True value of 39 + 42
8. True value of 29 × 11
9. Estimate of 29 × 11
10. The gap between them

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. True value of 612 – 388
12. Estimate of 612 – 388 to the nearest hundred
13. How far off was that estimate
14. True value of 4,812 + 3,190
15. How far off was the thousand estimate
16. Estimate of 486 + 317
17. True value of 486 + 317
18. The difference there
19. True value of 78 × 41
20. Estimate of 78 × 41

★ Challenge

NOT GRADED · REASON IT OUT

21. Rounding both numbers up always makes the estimate too high or too low — type high or low

22. Estimate 39 × 41, then give the true value

23. Rounding both numbers down makes the estimate — type high or low

24. Estimate 49 × 51, then give the true value

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

Big Numbers in Context

Millions stop being abstract when you attach them to something real.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up 11 ITEMS · RECALL

1 Zeros in one thousand

2 Zeros in one million

3 Thousands in a million

4 Hundreds in a thousand

5 Tens in a thousand

6 Zeros in ten thousand

7 Zeros in a hundred thousand

8 Ten thousands in a million

9 Hundreds in ten thousand

10 Tens in ten thousand

11 Thousands in ten thousand

Core GRADED · SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. Days in about a million minutes, rounded to the nearest hundred (A million minutes is about 694 days)

13. Dots per page 1,000 — pages for a million dots

14. Seconds in an hour

15. Hours in a week

16. Minutes in a day

17. Days in ten years, ignoring leap days

18. Seconds in a day

★ Challenge

NOT GRADED · REASON IT OUT

19. About how many years is a million days, to the nearest thousand

20. A stack of 100 sheets is 1 cm. Height in metres of a million sheets

21. About how many weeks is a million minutes, to the nearest whole

22. A million seconds in days, to the nearest whole

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

A Million Dots, Measured

Prove the estimate. Count one square, scale it up, defend the number.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

4 ITEMS · RECALL

1 Dots in a 10 by 10 square ______

2 Dots in ten such squares ______

3 Dots in a 5 by 5 square ______

4 Dots in four such squares ______

Core

GRADED · SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. Squares of 100 dots needed for a million ......

6. A 100 by 100 grid holds how many dots ......

7. Dots in a 20 by 20 square ......

★ Challenge

NOT GRADED · REASON IT OUT

8. How many 100 by 100 grids make a million

9. If one grid is 20 cm wide, the width in metres of 100 side by side

10. Dots in a 1,000 by 1,000 grid

11. Squares of 100 dots to reach 10,000

12. A page of 50 by 40 dots — pages to reach a million

13. Dots on 25 pages of 2,000

14. A 100 by 100 grid — grids needed for a million

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

TEACHER PAGE · ANSWER KEY

Week 2 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

2.1 · Rounding to Any Place

OUT OF 18

Warm-Up · 1–12 · 50 | 40 | 300 | 100 | 80 | 610 | 60 | 70 | 300 | 400 | 100 | 500

Core · 13–18 · 4800 | 5000 | 9000 | 5000 | 6000 | 7800

Challenge · 19–26 · 28000 | 100000 | 250 | 349 | 19000 | 46000 | 7500 | 8499

2.2 · Estimate Before You Compute

OUT OF 19

Warm-Up · 1–11 · 80 | 40 | 400 | 9 | 400 | 100 | 80 | 40 | 300 | 8 | 400

Core · 12–19 · 8000 | 200 | 1000 | 800 | 2200 | 8000 | 1000 | 3200

Challenge · 20–26 · 800 | 748 | 100 | 3072 | 500 | 240 | 3256

2.3 · How Wrong Is the Estimate

OUT OF 20

Warm-Up · 1–10 · 80 | 79 | 1 | 399 | 400 | 80 | 81 | 319 | 300 | 19

Core · 11–20 · 224 | 200 | 24 | 8002 | 2 | 800 | 803 | 3 | 3198 | 3200

Challenge · 21–24 · high | 1599 | low | 2499

2.4 · Big Numbers in Context

OUT OF 18

Warm-Up · 1–11 · 3 | 6 | 1000 | 10 | 100 | 4 | 5 | 100 | 100 | 1000 | 10

Core · 12–18 · 700 | 1000 | 3600 | 168 | 1440 | 3650 | 86400

Challenge · 19–22 · 3000 | 100 | 99 | 12

Fri · A Million Dots, Measured

OUT OF 8

Warm-Up · 1–4 · 100 | 1000 | 25 | 100

Core · 5–8 · 10000 | 10000 | 400 | 100

Challenge · 9–14 · 100 | 20 | 1000000 | 500 | 50000 | 100

NAME __________

DATE __________

RIGHT ____ OF ____ TRIED

Why Order Matters

Without an agreed order, one expression has several answers.

Warm-Up

12 ITEMS · RECALL

1 $2 + 3 \times 4$	2 $(2 + 3) \times 4$	3 $10 - 2 \times 3$	4 $(10 - 2) \times 3$
5 $6 + 6 \div 2$	6 $(6 + 6) \div 2$	7 $3 + 2 \times 5$	8 $(3 + 2) \times 5$
9 $12 - 3 \times 2$	10 $(12 - 3) \times 2$	11 $8 + 8 \div 4$	12 $(8 + 8) \div 4$

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $5 + 4 \times 3 - 2$ ....
14. $(5 + 4) \times (3 - 2)$ ....
15. $20 - 3 \times 4 + 2$ ....
16. $36 \div (2 + 4)$ ....
17. $36 \div 2 + 4$ ....
18. $20 - 3 \times 4 + 1$ ....
19. $(20 - 3) \times 4$ ....
20. $2 \times 6 + 3 \times 4$ ....
21. $2 \times (6 + 3) \times 4$ ....
22. $36 \div 6 \div 2$ ....

CHALLENGE AND THE ERROR JOURNAL ARE ON THE BACK

Challenge

NOT GRADED · REASON IT OUT

23. 2 × (3 + 4) × 5

24. Place brackets in 8 + 2 × 5 to make 50 — type the answer you get

25. 3 × (4 + 5) × 2

26. Brackets in 10 + 5 × 2 to make 30 — type the answer

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

Brackets First

Whatever is inside the brackets happens before anything else.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up
12 ITEMS · RECALL

1 (4 + 5) × 2	2 (9 – 3) ÷ 2	3 3 × (2 + 2)	4 (7 + 3) × 10
5 (12 – 4) ÷ 4	6 5 × (10 – 6)	7 (2 + 6) × 3	8 (15 – 5) ÷ 5
9 4 × (3 + 3)	10 (9 + 1) × 10	11 (20 – 8) ÷ 4	12 6 × (12 – 8)

Core
GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. (15 + 5) ÷ (2 + 2) ....
14. 4 × (6 + 3) – 10 ....
15. (8 × 3) – (4 × 2) ....
16. 2 × (50 – 15) ....
17. (25 – 9) ÷ (2 × 4) ....
18. ((3 + 3) × 4) – 10 ....
19. 100 ÷ (5 × 5) ....
20. (7 + 8) × (10 – 6) ....
21. (45 ÷ 9) + (6 × 3) ....

★ Challenge

NOT GRADED · REASON IT OUT

22. $100 - (25 \times 3)$

23. $((4 + 2) \times 3) - 8$

24. $120 \div (2 \times (3 + 3))$

25. $((10 + 2) \times 5) \div 3$

26. $240 \div (4 \times (2 + 1))$

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

NAME __________

DATE __________

RIGHT ____ OF ____ TRIED

Multi-Step Expressions

Multiply and divide before you add and subtract. Left to right within each.

Warm-Up

12 ITEMS · RECALL

1 $12 \div 3 + 2$	2 $12 + 3 \times 2$	3 $20 \div 4 \times 2$	4 $18 - 6 \div 3$
5 $7 \times 2 + 6$	6 $30 \div 5 - 2$	7 $18 \div 3 + 4$	8 $18 + 3 \times 4$
9 $24 \div 6 \times 2$	10 $20 - 8 \div 4$	11 $9 \times 3 + 3$	12 $40 \div 8 - 3$

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $48 \div 6 + 3 \times 4$ ....

14. $7 \times 8 - 6 \times 5$ ....

15. $64 \div 8 \div 2$ ....

16. $9 + 36 \div 4 - 3$ ....

17. $60 \div 5 + 7 \times 2$ ....

18. $81 \div 9 - 3 \times 2$ ....

19. $7 \times 8 + 100 \div 4$ ....

★ Challenge

NOT GRADED · REASON IT OUT

20. 100 − 4 × 12

21. 144 ÷ 12 + 8 × 3 − 5

22. 2 × 3 + 4 × 5 + 6 × 7

23. 11 × 4 − 22 ÷ 2

24. 144 ÷ 12 × 3

25. 200 ÷ 8 + 6 × 7 − 9

26. 3 × 4 + 5 × 6 + 7 × 8

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

Write the Expression

Turn the sentence into symbols, then work it out.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

12 ITEMS · RECALL

1 Five more than three lots of four

2 Double seven, then add one

3 Ten less than six times three

4 Half of twenty, plus four

5 Three lots of nine

6 Twenty shared by four

7 Four more than double six

8 Triple five, then add two

9 Twenty less than five times six

10 A third of thirty, plus one

11 Two lots of nine, minus eight

12 Half of sixteen, times three

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. The sum of 8 and 4, multiplied by 3

14. 8 added to 4 lots of 3

15. 100 less the product of 7 and 9

16. The difference of 20 and 8, divided by 4

17. Six lots of the sum of 2 and 3

18. Six times the sum of two and five

19. The product of 7 and 8, less 20

20. Divide 96 by the sum of 4 and 8

21. Add 15 to the product of 6 and 9

22. Double the difference between 30 and 12

★ Challenge

NOT GRADED · REASON IT OUT

23. Three times the difference between 15 and 7

24. The sum of 12 and 8, divided by the difference of 9 and 4

25. Four times the difference between 25 and 13

26. The sum of 30 and 20, divided by the difference of 12 and 7

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

4 ITEMS · RECALL

1 Round 4,829 to the nearest hundred

2 2 + 3 × 4

3 Round 3,451 to the nearest hundred

4 5 + 2 × 3

Core

GRADED · SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. (5 + 4) × (3 – 2)

6. 48 ÷ 6 + 3 × 4

7. Zeros in one million

8. The sum of 8 and 4, multiplied by 3

9. (20 – 3) × 4

10. Estimate 396 × 21

★ Challenge

NOT GRADED · REASON IT OUT

11. Estimate 187 × 4

12. 144 ÷ 12 + 8 × 3 − 5

13. 5 + 4 × 3 − 12 ÷ 4

14. Largest whole number that rounds to 700 at the nearest hundred

15. Seconds in a day

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

TEACHER PAGE · ANSWER KEY

Week 3 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

3.1 · Why Order Matters

OUT OF 22

Warm-Up · 1–12 · 14 | 20 | 4 | 24 | 9 | 6 | 13 | 25 | 6 | 18 | 10 | 4

Core · 13–22 · 15 | 9 | 10 | 6 | 22 | 9 | 68 | 24 | 72 | 3

Challenge · 23–26 · 70 | 50 | 54 | 30

3.2 · Brackets First

OUT OF 21

Warm-Up · 1–12 · 18 | 3 | 12 | 100 | 2 | 20 | 24 | 2 | 24 | 100 | 3 | 24

Core · 13–21 · 5 | 26 | 16 | 70 | 2 | 14 | 4 | 60 | 23

Challenge · 22–26 · 25 | 10 | 10 | 20 | 20

3.3 · Multi-Step Expressions

OUT OF 19

Warm-Up · 1–12 · 6 | 18 | 10 | 16 | 20 | 4 | 10 | 30 | 8 | 18 | 30 | 2

Core · 13–19 · 20 | 26 | 4 | 15 | 26 | 3 | 81

Challenge · 20–26 · 52 | 31 | 68 | 33 | 36 | 58 | 98

3.4 · Write the Expression

OUT OF 22

Warm-Up · 1–12 · 17 | 15 | 8 | 14 | 27 | 5 | 16 | 17 | 10 | 11 | 10 | 24

Core · 13–22 · 36 | 20 | 37 | 3 | 30 | 42 | 36 | 8 | 69 | 36

Challenge · 23–26 · 24 | 4 | 48 | 10

Fri · Mid-Unit Quiz

OUT OF 11

Warm-Up · 1–4 · 4800 | 14 | 3500 | 11

Core · 5–11 · 9 | 20 | 800 | 6 | 36 | 68 | 8000

Challenge · 12–15 · 31 | 14 | 749 | 86400

NAME __________

DATE __________

RIGHT ____ OF ____ TRIED

The Little Raised Number

An exponent counts how many times the base is multiplied by itself.

Warm-Up

10 ITEMS · RECALL

1 2^2	2 3^2	3 5^2	4 2^3
5 4^2	6 6^2	7 7^2	8 8^2
9 9^2	10 1^5		

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. 3^3 ....
12. 2^4 ....
13. 5^3 ....
14. 2^5 ....
15. 4^3 ....
16. 11^2 ....

Challenge

NOT GRADED · REASON IT OUT

17. 10^2

18. 10^3

19. 2^8

20. Which is bigger, 3^4 or 4^3 — type the value of the bigger

21. 2^{10}

22. Which is bigger, 2^6 or 6^2 — type the value of the bigger

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

NAME __________

DATE __________

RIGHT ____ OF ____ TRIED

Powers of Ten

The exponent is the number of zeros. That is the whole shortcut.

Warm-Up

3 ITEMS · RECALL

1 10^1

2 Zeros in 10^7

3 10^0

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

4. Write 5,000 as a digit times a power of ten — type the exponent

★ Challenge

NOT GRADED · REASON IT OUT

5. 10^2
6. 10^3
7. 10^4
8. Zeros in 10^5
9. 10^6
10. 3×10^2
11. 7×10^3
12. 4×10^4
13. 25×10^2
14. $10^3 \times 10^2$
15. A million written as a power of ten — type the exponent
16. 10^5

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

Expanded Form with Powers

Every place value is a power of ten in disguise.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

12 ITEMS · RECALL

1 The value of the 4 in 400

2 The value of the 7 in 70

3 The value of the 3 in 3,000

4 200 + 30 + 5

5 4,000 + 200 + 10

6 The value of the 9 in 900

7 How many hundreds make 2,600

8 How many hundreds make 4,300

9 How many hundreds make 7,100

10 How many hundreds make 9,500

11 How many tens make 3,000

12 How many tens make 6,000

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

20. In 4,271 — what is the 4 worth

21. In 4,271 — what is the 2 worth

22. In 4,271 — what is the 7 worth

23. In 8,605 — what is the 8 worth

24. In 8,605 — what is the 6 worth

25. In 3,940 — what is the 9 worth

26. In 7,128 — what is the 7 worth

27. In 7,128 — what is the 1 worth

28. In 5,083 — what is the 5 worth

29. Bigger: 3,421 or 3,412

30. Bigger: 5,680 or 5,806

★ Challenge

NOT GRADED · REASON IT OUT

33. $6 \times 10^3 + 4 \times 10^2 + 2$

34. $3 \times 10^4 + 5 \times 10^2$

35. The value of the 8 in 84,000

36. $9 \times 10^2 + 9 \times 10^1 + 9$

37. 7×10^5

38. $2 \times 10^5 + 6 \times 10^3 + 4 \times 10^1$

39. The largest power of ten inside 47,300 — type its exponent

40. $5 \times 10^3 + 2 \times 10^2$

41. $7 \times 10^4 + 3 \times 10^2$

42. $4 \times 10^3 + 6 \times 10^1 + 2$

43. 20,000 + 4,000 + 300 + 50

44. $3 \times 10^5 + 7 \times 10^3 + 5 \times 10^1$

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

Error Journal Sweep

Mixed review of the whole mission. Fix only what repeats.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up
10 ITEMS · RECALL

1 Round 149 to the nearest hundred

2 $2 + 3 \times 4$

3 Zeros in one million

4 5^2

5 $(2 + 3) \times 4$

6 Round 340 to the nearest hundred

7 $5 + 2 \times 3$

8 Zeros in ten thousand

9 6^2

10 $(3 + 2) \times 5$

Core
GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. $48 \div 6 + 3 \times 4$

12. Estimate $3{,}964 \div 4$

13. Estimate $486 + 317$ to the nearest hundred

14. $60 \div 5 + 7 \times 2$

15. 2^4

CHALLENGE AND THE ERROR JOURNAL ARE ON THE BACK

★ Challenge

NOT GRADED · REASON IT OUT

16. 10^3

17. Round 96,412 to the nearest ten thousand

18. 4×10^4

19. $6 \times 10^3 + 4 \times 10^2 + 2$

20. $144 \div 12 + 8 \times 3 - 5$

21. 2^8

22. 10^4

23. Round 24,650 to the nearest thousand

24. $4 \times 10^3 + 6 \times 10^1 + 2$

25. $200 \div 8 + 6 \times 7 - 9$

26. 2^{10}

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

Mission 03 Test

Twelve items plus the Big Question, answered out loud.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

4 ITEMS · RECALL

1 Round 612 to the nearest ten

2 3^2

3 Round 448 to the nearest ten

4 4^2

Core

GRADED · SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. Round 4,829 to the nearest thousand

6. $10 - 2 \times 3$

7. $(15 + 5) \div (2 + 2)$

8. 2^5

9. Estimate 4,812 + 3,190 to the nearest thousand

10. The sum of 12 and 8, divided by the difference of 9 and 4

11. $(15 - 5) \div 5$

CHALLENGE AND THE ERROR JOURNAL ARE ON THE BACK

★ Challenge

NOT GRADED · REASON IT OUT

12. $100 - 4 \times 12$

13. 7×10^3

14. $2 \times 10^5 + 6 \times 10^3 + 4 \times 10^1$

15. $2 \times 3 + 4 \times 5 + 6 \times 7$

16. 7×10^4

17. $3 \times 4 + 5 \times 6 + 7 \times 8$

18. A million seconds in days, to the nearest whole

19. 5^3

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

TEACHER PAGE · ANSWER KEY

Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

4.1 · The Little Raised Number

OUT OF 16

Warm-Up · 1–10 · 4 | 9 | 25 | 8 | 16 | 36 | 49 | 64 | 81 | 1

Core · 11–16 · 27 | 16 | 125 | 32 | 64 | 121

Challenge · 17–22 · 100 | 1000 | 256 | 81 | 1024 | 64

4.2 · Powers of Ten

OUT OF 4

Warm-Up · 1–3 · 10 | 7 | 1

Core · 4 · 3

Challenge · 5–16 · 100 | 1000 | 10000 | 5 | 1000000 | 300 | 7000 | 40000 | 2500 | 100000 | 6 | 100000

4.3 · Expanded Form with Powers

OUT OF 23

Warm-Up · 1–12 · 400 | 70 | 3000 | 235 | 4210 | 900 | 26 | 43 | 71 | 95 | 300 | 600

Core · 20–30 · 4000 | 200 | 70 | 8000 | 600 | 900 | 7000 | 100 | 5000 | 3,421 | 5,806

Challenge · 33–44 · 6402 | 30500 | 80000 | 999 | 700000 | 206040 | 4 | 5200 | 70300 | 4062 | 24350 | 307050

Thu · Error Journal Sweep

OUT OF 22

Warm-Up · 1–12 · 100 | 14 | 1000 | 6 | 25 | 20 | 300 | 11 | 10000 | 4 | 36 | 25

Core · 13–22 · 100000 | 20 | 40000 | 1000 | 6402 | 800 | 26 | 25000 | 4062 | 16

Challenge · 23–26 · 31 | 256 | 58 | 1024

Fri · Mission 03 Test

OUT OF 14

Warm-Up · 1–4 · 610 | 9 | 450 | 16

Core · 5–14 · 5000 | 4 | 5 | 52 | 7000 | 32 | 8000 | 4 | 2 | 70000

Challenge · 15–19 · 206040 | 68 | 98 | 12 | 125